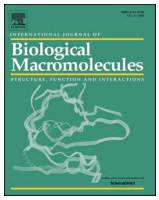
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Thermodynamics insights of lignin dissolution in deep eutectic solvents

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ABSTRACT

To develop a green solvent for lignin dissolution, the most fundamental aspects of its mechanism must be elucidated. Understanding the thermodynamic behaviors is of significant importance for designing novel deep eutectic solvents (DESs) to dissolve lignin. Herein, the heat of dissolution of lignin in acidic, alkaline or neutral DESs was determined by high-precision solution microcalorimetry, and comprehensively investigated the effect of physicochemical properties of DESs on the heat of dissolution and solubility of lignin. The resulted observed that lignin dissolution in alkaline DESs was more vigorous exothermic than that in acidic and neutral DESs. The thermodynamic parameters for the dissolution of lignin in the DESs were calculated from the heat of dissolution and solubility. The positive free energy change indicated that the process of lignin dissolution was non-spontaneous and unfavorable thermodynamically. By analyzing the changes of enthalpy and entropy of lignin solution systematically, lignin dissolution in DESs was enthalpically favorable and controlled by the entropy term. Especially, the interaction entropy predominantly made the free energy change unfavorable for lignin dissolution. The achieved thermodynamic findings could serve as the basis to construct robust DESs for lignin biorefinery.

1. Introduction

Lignin, a unique renewable bioresource with significant aromatic characteristics, has great potential of serving as raw material for the production of functionalized aromatic compounds and alternatives of petroleum-based products [1]. Structurally, lignin is a complex heterogeneous biomacromolecule composed of three primary phenylpropane units, *i.e.*, *p*-coumaryl, coniferyl, and sinapyl, which are firmly bonded together by various types of ether bonds (*e.g.*, β-O-4, α-O-4, and 4-O-5) and carbon-carbon bonds (*e.g.*, β-β, β-5, β-1, and 5–5) (Fig. 1) [2,3]. To better adjust the properties of lignin, it is essential to dissolve lignin efficiently and recover the processed lignin from the solution controllably.

As a promising class of green and designable solvents, deep eutectic solvents (DESs) have been extensively applied in the fractionation of lignocellulosic biomass [4–7]. Selective extraction of lignin or hemicellulose by DESs can be realized owing to the excellent ability of DESs to dissolve lignin or hemicellulose selectively. Besides, various structurally complex technical lignin can also be efficiently dissolved in a variety of designed DESs, which can be divided into three types: acidic, alkaline, and neutral [8–10]. Lignin dissolution in DESs is primarily due to the rich hydrogen-bonding network structure present in DESs, enabling strong interactions with lignin macromolecular chains, and dissolution is often accompanied by depolymerization and polymerization reactions of lignin [11]. As for acidic DESs, which effectively breaking the labile ether bonds (*e.g.* *β*-O-4) of lignin to form reactive benzylic carbocations during dissolution, which in turn inducing condensation reactions to form recalcitrant carbon-carbon bonds in lignin [4]. Compared with acidic DESs, alkaline DESs can cleave the *β*-O-4 bonds in non-phenolic units result in the formation of epoxide intermediates and the *β*-O-4 bonds in phenolic units lead to the formation of a quinone-methyl type intermediates, respectively [12]. Generally, dissolution processes are governed by both kinetics and thermodynamics [13]. Among them, thermodynamics has been recognized as an effective tool for the macroscopic study of the dissolution processes, and this approach has been used to study the dissolution of biopolymers, *i.e.*,

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starch, cellulose and lignocellulose [14–16]. Regarding the dissolution energetics of polymers, most of the information for the specific polymer-solvent interactions could be obtained either from molecular dynamics simulations [17–20] or the analysis of solvatochromic parameters [8,16]. For lignin dissolution in DESs, it is highly important to understand the energetics for the interactions between lignin and DESs. Unfortunately, the thermodynamics behaviors of lignin dissolution in DESs are rarely investigated specifically.

From the perspective of thermodynamics, the dissolution of lignin in neat DESs is probably controlled by the changes of Gibbs free energy, whose thermodynamic equation is consisted of enthalpy changes and entropy changes terms. Theoretical approaches have revealed that the solubility of polymers in a given solvent is fundamentally dependent on the enthalpy changes and entropy changes during dissolution. The enthalpy changes can be precisely obtained by carefully determining the heat during the dissolution of lignin in DESs under the constant temperature and pressure. However, it was difficult to achieve entropy changes by traditional models, such as Flory-Huggins mean-field theory and regular solution theory [21] because they ignored the entropy changes caused by the special interactions between polymer and solvent during the dissolution processes. Generally, the dissolution of polymers was always associated with small positive entropy changes resulted from the mixing of polymers and solvent [22], combined with the fact that the dissolution of cellulose and lignin in ionic liquids was the exothermic processes [15,23]. Nevertheless, it could not deduce whether the dissolution of lignin in DESs was spontaneous or not by only considering the mixing entropy changes. Therefore, much effort should be devoted to precisely understand the complex dissolution process of lignin in DESs from the thermodynamics viewpoint.

The number of hydroxyl, amino, and carboxyl groups in the hydrogen bond donors (HBDs) and hydrogen bond acceptors (HBAs) of DESs determines the characteristics of the hydrogen bond network of DESs and the strength of the hydrogen bonding interactions with lignin. Therefore, we prepared a series of DESs for lignin dissolution by selecting different HBAs and HBDs with varied functional groups (Fig. 1 and Table S1). Enzymatic hydrolysis lignin produced by the second-generation bioethanol industry was a high-value, large-scale byproduct. Compared to other technical lignins (*e.g.* alkali lignin, kraft lignin), enzymatic hydrolysis lignin was extracted under relatively mild conditions, with a low degree of structural disruption as well as condensation, thus better reflecting the structure and properties of natural lignin [24]. In this work, we have experimentally determined for the first time the heat of dissolution of lignin in various DESs and calculated the thermodynamic parameters of lignin dissolution in DESs in conjunction with lignin solubility. The obtained results suggested that the properties of the used DESs significantly affected the thermodynamics behaviors of lignin dissolution. Notably, the dissolution of lignin in DESs was a violently exothermic process, and the exothermic values for alkaline DESs were much higher than those for both acidic and neutral DESs. The stronger interactions between the lignin macromolecular chains and DESs molecules at the dissolution state resulted in a highly negative entropy change, and the interaction between lignin and alkaline DESs were much stronger than that in acidic and neutral DESs. Therefore, lignin dissolution in DESs was enthalpically favorable and controlled by the entropy term. These thermodynamic insights could better understand the lignin solvation mechanisms in DESs and the interactions between lignin and DESs components, which allow the development of more effective DESs for lignin dissolution, extraction, and processing.

2. Materials and methods

2.1. Materials

Choline chloride (ChCl), proline (Pro), betaine (Bet), urea (U), ethanolamine (EA), *N*-methylthiourea (NMTU), ethylene glycol (EG), glycerol (Gly), lactic acid (LA), and oxalic acid (OA) were purchased from Macklin Chemical Co., Ltd. (Shanghai, China). Allyltrimethylammonium chloride (ATMAC), formic acid (FA), acetic acid (AA), and propanoic acid (PA) were purchased from Aladdin Biochemical Technology Co., Ltd. Enzymatically hydrolyzed lignin was selected as the lignin sample for this investigation, which was extracted from corncobs and supplied by Shandong Longlive Bio-Technology Co., Ltd.

2.2. Determination of lignin solubility and dissolution heat

First, all DESs were prepared by mixing the HBDs and HBAs at a molar ratio of 1:2, and stirred at 70 °C to obtain a clear and transparent solvent (Fig. 1 and Table S1). Then, the solubility of lignin in the developed DESs was determined, and the procedure was referred to the previous literature [8,16]. The specific experimental details were provided in Supporting Information.

The heat of dissolving lignin in one DES was measured using a high-precision solution microcalorimeter (C80, Setaram Instruments). The calorimeter vessel was filled with DESs (1 g) and with a sample cell containing about 10 mg of lignin. The calorimeter vessel was heated to 70 °C at a rate of 1 °C min^{-1} and then hold for thermal equilibration.

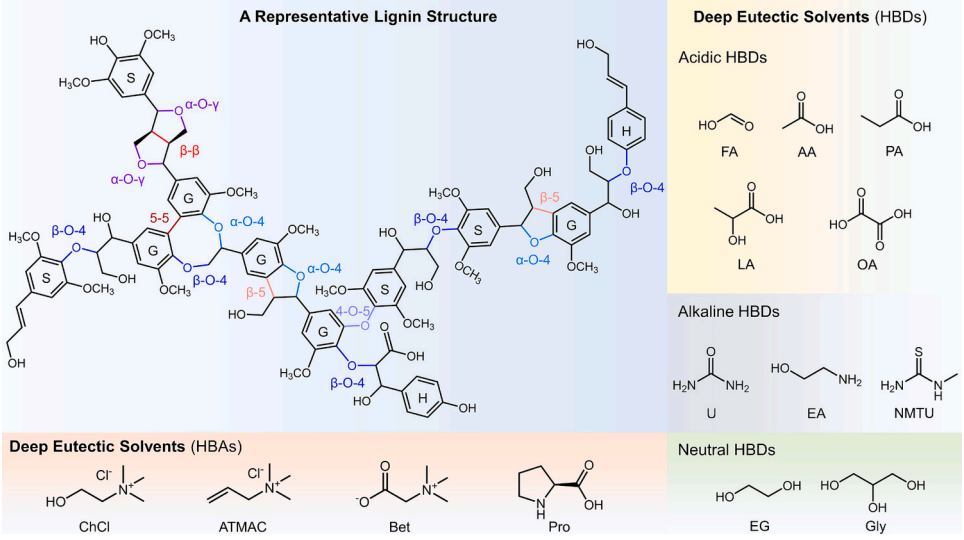


Fig. 1. Chemical structures of lignin and the prepared DESs.

Lignin was mixed with DESs when the heat flow was constant and stirred at 500 rpm. Subsequently, the dissolution of lignin induced a rapid change in heat flow, which was integrated into the heat of dissolving lignin in DESs.

2.3. Characterization of DESs

The Kamlet-Taft (K-T) solvatochromic parameters of DESs were determined according to the procedure reported previously [16,25]. The detailed procedures and calculation equations were given in the Supporting Information. The pH values of DESs were determined from an aqueous solution (2 mol·L^{-1}) of each DES, and the pH values were measured by the pH-meter (PHS-3C, Shanghai INESA Scientific Instrument Co., Ltd., China) at 25 °C. The reported pH values are average of three independent measurements. The density of various DESs were measured using an Anton Paar DMA 5000 M density meter. The ionic conductivity (σ) of the DESs was determined using a conductivity instrument (DDS-307 A, INESA). The surface tensions of DESs samples were measured by an automated tensiometer JY-82B (Chengde Dingsheng Instrument Co., Ltd.) with Du Nouy ring method.

2.4. Structural characterization of lignin

The FT-IR spectra of enzymatic hydrolysis lignin was collected on a Tensor II spectrometer (Bruker, Germany) in the frequency range of 400–4000 cm^{-1} using the potassium bromide pellet technique. The weight average molecular weight (M_w) of the enzymatic hydrolysis lignin was estimated with gel permeation chromatography (GPC) (Agilent 1200, Japan), and lignin was fully acetylated in acetic anhydride/pyridine before GPC analysis. Acetylated lignin (4 mg) was dissolved in 2 mL of HPLC-grade tetrahydrofuran and then 10 μL solution was injected into vial for testing. NMR spectra were recorded on an Avance 400 MHz spectrometer (Bruker, Germany). The ^{31}P NMR and 2D HSQC NMR analyses of lignin were referred to the previous operation [24]. For quantitative ^{31}P NMR, 20 mg lignin was dissolved in 0.5 mL of pyridine/$CDCl_3$ (1.6/1, *v*/v) and subsequently phosphorylated with 2-chloro-4,4,5,5-tetramethyl-1,3,2-dioxaphospholane. For 2D HSQC NMR, 50 mg of lignin was dissolved in 0.5 mL of DMSO-d_6. The spectral widths were 5000 and 20,000 Hz for the ^1H and ^{13}C dimensions, respectively. A total of 1024 points were collected for the ^1H dimension with a recycle delay of 1.5 s, while 64 or 256 transients were used for the ^{13}C dimension. The semi-quantitative analyses were conducted to calculate the relative contents of lignin interlinkage using the integration of α-C/H correlation peaks in the HSQC [26], and the assignments of $^{13}C—^1H$ correlated signals in the HSQC spectra of lignin were listed in Table S2. The relative content of the lignin interlinkages was given as the amount per 100 aromatic units, and the relative content was calculated from the integrals as follows:

$$I(C_9\ \text{units}) = 0.5I(S_{2,6}) + I(G_2) + 0.5I(H_{2,6}) \quad (1)$$

$$\beta - O - 4\ \text{content (per 100 Ar)} = I(\beta - O - 4_\alpha)/I(C_9) \times 100 \quad (2)$$

Where $I(S_{2,6})$, $I(G_2)$, and $I(H_{2,6})$ were the integral values of $S_{2,6}$, G_2, and $H_{2,6}$, respectively. $I(C_9$ units) represented the integral value of the aromatic ring, and I(β-O-4α) was the integral value of the α-position of β-O-4.

3. Results and discussions

3.1. Study on dissolution heat of lignin

Solution microcalorimeter was used to determine the heat absorbed or released in the process of dissolving lignin into DESs. On the solution microcalorimeter, the initial state and the final state of the lignin-DES system could be well defined, which was more precise than traditional differential scanning calorimetry determination because the initial state of the polymer-solvent could not be determined, resulting from the physical transformation of the polymer in the mixing process [14,27]. For all the experiments of determining the dissolution heat of lignin at 70 °C, the amounts of lignin and DES were 10 mg and 1 g, respectively, to ensure that the amount of dissolved lignin was within the maximum solubility of lignin in the prepared DESs (Table S3). The heat of dissolution of lignin in different DESs systems could be counted according to the measured heat flows (Fig. S1). It was observed that dissolution heat of lignin in all studied DESs were negative (Fig. 2). Notably, ChCl-EA showed the highest dissolution heat (−340.737 J·g^{-1}), while Pro-LA provided the lowest dissolution heat (−9.873 J·g^{-1}). These results confirmed that the dissolution of lignin was an exothermic process, implying that the released energy for the formation of lignin-DES interactions could well compensate the energy loss for the breaking of the DES-DES and lignin-lignin interactions.

Structural features of lignin could provide an effective understanding of changes in interactions during lignin dissolution in DESs. The ^{31}P NMR was used to characterize and quantify the different hydroxyl groups in the lignin. The signals of aliphatic hydroxyl (-OH) and carboxyl (-COOH) groups in lignin were detected at 145.5–149.0 and 134.0–136.0 ppm, respectively. The peaks of C_5-substituted (including syringyl and condensed), guaiacyl, and *p*-hydroxyphenyl phenolic hydroxyl groups were detected at 141.8–143.3, 138.7–140.2, and 137.2–138.5 ppm, respectively (Fig. 3a) [28]. The different amounts of OH groups in the lignin listed in Table S4. As detected, the aliphatic OH, phenolic OH, and carboxylic acid OH contents were 1.20, 2.05, and 0.98 mmol·g^{-1}, respectively. The abundant hydroxyl groups of lignin provided a large number of hydrogen bond interaction sites for dissolution in DESs [2], and a large amount of heat was released from the hydrogen bond interactions formed as the lignin dissolution in DESs. The basic functional group of lignin was captured by FT-IR spectra. As shown in Fig. S2, a wide absorption band of -OH stretching (3425 cm^{-1}) and the typical aromatic ring skeleton stretching signals (1600, 1515, and 1462 cm^{-1}) were observed for lignin. The peaks at 1265 and 1125 cm^{-1} were attributed to the stretching of the guaiacyl and syringyl unit, respectively, and the signal at 835 cm^{-1} represented C—H stretching in the *p*-hydroxyphenyl unit [12]. In addition, the signal at 1700 cm^{-1} was attributed to the stretching vibrations of unconjugated carbonyl group [10], which was also an effective site for the formation of hydrogen bond interactions with DESs. Lignin dissolution was accompanied by the cleavage of the interlinkages of the basic aromatic units, such as α-aryl, β-aryl, and α-alkyl ether bonds [2,29]. 2D HSQC NMR spectrum were conducted to reveal lignin corresponding substructures (Fig. 3b). Enzymatic hydrolysis lignin showed obvious cross-signals that assigned to substructures of β-O-4 linkages (A) for side chain. The correlated signal of α-position in the side chain of β-O-4 linkages were detected at δ_C/δ_H 71.9/4.86, whereas the correlated peak at δ_C/δ_H 86.0/4.12 was assigned to their β-position in the side chain linked to syringyl units in β-O-4 linkages and the peak at δ_C/δ_H 59.8/3.38–3.80 corresponded to

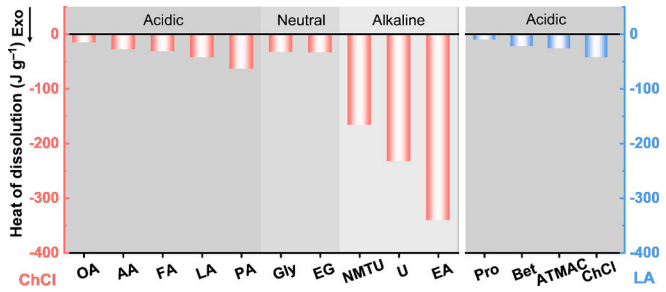


Fig. 2. Dissolution heat of lignin in various DESs at 70 °C. The red and blue bars were for ChCl-based and LA-based DESs, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

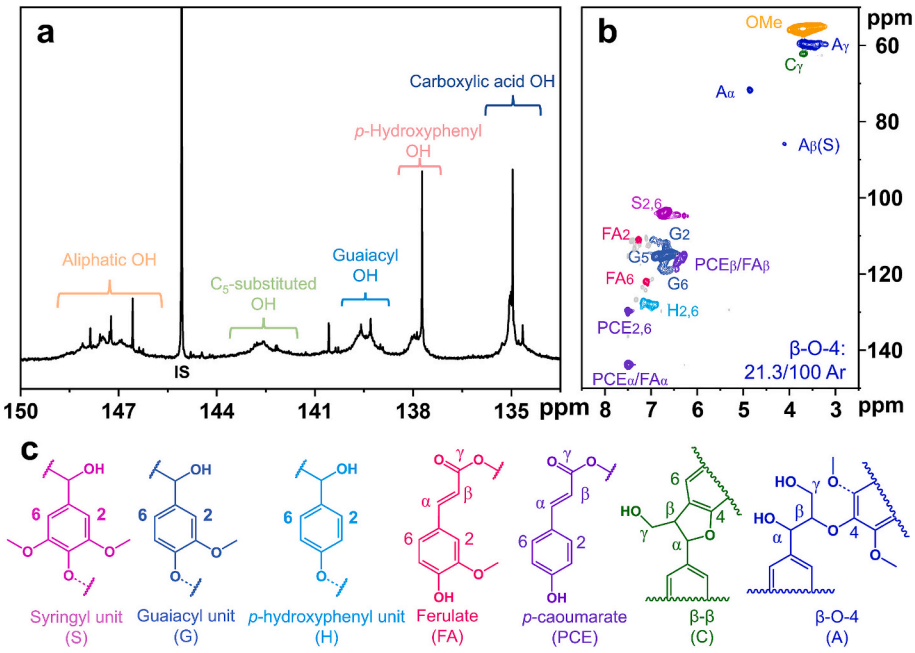


Fig. 3. ^{31}P NMR spectrum (a) and 2D HSQC NMR spectrum (b) of enzymatic hydrolysis lignin, and the main substructures of lignin (c).

the γ-position of β-O-4 linkages [24]. The content of β-O-4 linkage was semi-quantitatively calculated by HSQC, and the content of dominant β-O-4 in enzymatic hydrolysis lignin was 21.3/100 Ar. Structural properties of regenerated lignin from acidic, neutral, and alkaline DESs were evaluated by HSQC NMR (Fig. S3). These regenerated lignin samples exhibited almost identical cross-peaks compared with original lignin (Fig. 3b), which indicated that the DESs were less effective in cleavage the lignin interlinkages during the dissolution process. Specifically, the β-O-4 content of lignin regenerated from acidic (ChCl-OA), neutral (ChCl-EG), and alkaline (ChCl-EA) DESs decreased slightly to 16.4, 19.6, and 17.5/100 Ar, respectively. The lignin dissolution process required the absorption of energy to cleave the β-aryl ether bond, exhibiting an endothermic process. Nevertheless, DESs showed weak cleavage effect on lignin β-aryl ether bond. Therefore, the dissolution of lignin in DES was a violently exothermic process, suggesting that the exothermic processes exhibited by the hydrogen bond interactions between the abundant hydroxyl and carbonyl functional groups on lignin and DES were much larger than the endothermic processes of weak cleavage of the lignin β-aryl ether bond. In addition, the cleavage of β-aryl ether bonds further increased the phenolic hydroxyl content, thus providing more hydrogen bond sites to facilitate heat release.

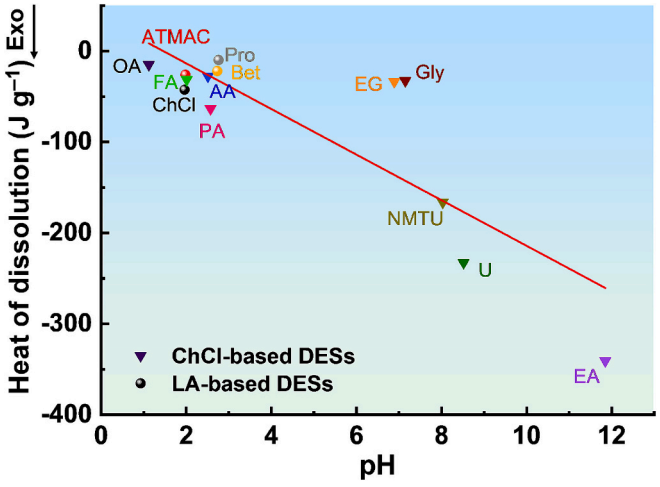


Fig. 4. The linear correlation between the dissolution heat and pH values of the studied DESs.

3.2. *Effect of acid-base properties and hydrogen bond acidity-basicity of DESs in dissolution heat*

Noteworthily, the dissolution heat significantly varied with the acid-base properties of DESs, and the exothermic values for alkaline DESs were much higher than those for both acidic and neutral DESs (Fig. 2). Furthermore, the correlation between the dissolution heat and the pH values of DESs was evaluated, and an approximate linear relationship was observed (Fig. 4). For ChCl-EA with a pH of 11.85, the heat dissolution (−340.737 J·g^{-1}) was about 10 and 23 times more exothermic than that for ChCl-EG (−33.606 J·g^{-1}) with a pH of 6.89 and ChCl-OA (−15.129 J·g^{-1}) with a pH of 1.12, respectively. Besides, the released dissolution heat decreased with the decrease of the pH values, indicating that the amount of the formed lignin-DES interactions might be less. Generally, the heat released was related to the number of solvent molecules bound to the lignin macromolecular chains during dissolution [17,30]. In the lignin structure, phenolic OH was the primary OH group and accounted for 48.5 % of whole OH groups (Table S4), which could form more solvent molecules-lignin interactions in alkaline environment through base-driven deprotonation [29,31]. Therefore, the higher exothermicity of lignin dissolution in alkaline DESs compared to acidic and neutral DESs was mainly driven by the strong base-induced deprotonation of the abundant phenolic OH of lignin and the formation of more solvent molecules-lignin interaction, which released significant heat because of these chemical reactions and interactions. Nonetheless, no correlation could be directly achieved between lignin solubility and the heat of dissolution of lignin in DESs. For example, the solubility of lignin in the ChCl-U was only 52 mg·g^{-1} although the heat of dissolution of lignin in ChCl-U was relatively high (−232.797 J·g^{-1}). This phenomenon could be interpreted by the fact that the lignin dissolution was a complicated process, which could be affected not only by thermodynamic, but also by the kinetic factors such as the penetration of DESs molecules and the chain depolymerization of lignin [32]. The effect of physical properties of DESs in lignin solubility was specifically analyzed below.

As reported, Kamlet-Taft parameters can be utilized to assess the ability of DESs to dissolve lignin, which can be categorized primarily into hydrogen bond acidity (α) and basicity (β). The DESs with higher α or β values had stronger ability to provide or accept hydrogen bonding, which were beneficial for breaking the intramolecular hydrogen bonds in lignin to form stronger DES-lignin interactions, thereby increasing the solubility of lignin [8,33,34]. Thereby, the correlation between the dissolution heat of lignin and the Kamlet-Taft parameters (α and β) were subsequently investigated. After carefully analyzing the determined Kamlet-Taft parameters (Table S5) and the dissolution heat of lignin in DESs (Fig. 2), it was observed that there were no definite correlations between the α/β values and the dissolution heat of lignin (Fig. 5). For instance, the dissolution heat (-15.129 J·g^{-1}) was less exothermic in the ChCl-OA although it had strong ability to form hydrogen bonds with lignin according to its Kamlet-Taft parameters ($\alpha = 2.032$, and $\beta = 0.925$). In comparison, the α and β values of ChCl-U were only 0.926 and 0.497, respectively, but lignin dissolution in ChCl-U was more exothermic (-232.797 J·g^{-1}). Notably, the dissolution heat of lignin sharply shifted to less exothermic (-21.989 J·g^{-1}) in Bet-LA with similar α (0.979) and β (0.519) values as ChCl-U. These results implied that the hydrogen-bond providing and accepting capacity of DESs (Kamlet-Taft parameters) might not be a very accurate indictor to evaluate the exothermic process of forming DESs-lignin interactions although they could roughly indicate the strength of the interactions between DESs and lignin.

3.3. Effect of physical properties of DESs in lignin solubility

Based on the above discussion, although alkaline DES molecules were more likely to form hydrogen bond interactions and release more heat than acidic and neutral DES molecules when quantitative lignin dissolution, it did not mean that alkaline DES can dissolve more lignin. The lignin dissolution in DESs was exothermal, implying that an increase in temperature should not thermodynamically favor the dissolution process. However, the lignin solubility increased with an increase in the temperature from 50 to 70 °C for all used DESs (Table S3). Considering that all DESs showed satisfactory ability to dissolve the lignin at 70 °C, we selected 70 °C as the suitable temperature for the dissolution experiments. To gain insight into the kinetics of the lignin dissolution and to reveal the potential reasons limiting further lignin dissolution, the ionic nature and surface tension of the studied DESs were determined. It could be observed that DESs with higher conductivity (> 2000 uS·cm^{-1}) generally exhibited a higher lignin solubility (Table S3, S5). For ChCl-EG with a conductivity of 6854 uS·cm^{-1}, the lignin solubility at 70 °C (0.685 g·g^{-1}) was about 2.6 times more dissolved than that for ChCl-Gly (0.258 g·g^{-1}) with a conductivity of 1142 uS·cm^{-1}. The high conductivity of DESs could provide more flexible ion mobility during the dissolution, accelerating the formation of hydrogen bonding interactions between lignin macromolecular chains and DESs molecules. In addition, the conductivity of the DESs could be increased, resulting in a more uniform distribution of hydrogen bonding interactions formed through molecular collisions [11]. DESs surface tension depended on the strength of intermolecular interactions between HBAs and HBDs and played a key role in the mass transfer process of dissolution. Surface tension was an important physical property which has significant effects on permeability. In general, the lower the surface tension of DESs, the stronger the droplet formation ability and the easier it was to penetrate into the narrow space inside the macromolecule [35,36]. The high values of surface tension for ChCl-Gly (57.3 mN·m^{-1}), ChCl-U (53.6 mN·m^{-1}), and ChCl-OA (60.6 mN·m^{-1}) were due to the higher number of hydrogen bonding for the same molar ratios of DESs (Table S5). In these DESs hydrogen bonding networks, there were not only interactions between HBAs and HBDs, but also the dense hydrogen bonding network formed by the HBDs itself, which could seriously hinder the formation of hydrogen bonding interactions between lignin macromolecular chains. In conclusion, conductivity and surface tension characterizations provided important physicochemical parameters for exploring the kinetics of lignin dissolution. Higher conductivity might accelerate the formation of DESs-lignin interactions, while lower surface tension facilitated DESs penetration into the lignin complex structure.

3.4. Study on the thermodynamic parameters of lignin dissolution

The thermodynamic parameters for the dissolution of lignin in the DESs were further studied. According to the heat flow profiles of different quantity of dissolved lignin in specific DESs (ChCl-EG), the released total heat (J) was approximately proportional to the quantity of dissolved lignin (Fig. S4a). Notably, the heat of dissolution for dissolving 4, 6, 8, and 10 mg of lignin in specific DESs (ChCl-EG) were − 38.507, −36.167, −34.329, and − 33.606 J·g^{-1}, respectively (Fig. S4b), which indicated that specific DESs dissolved different quantity of lignin with similar heat of dissolution. These thermodynamic results were very similar with those for the dissolution of microcrystalline cellulose in ionic liquids [15]. From the corresponding heat of dissolution (H_{dis}) and the actual quantity of dissolved lignin in DESs at 70 °C (m), the enthalpy change (ΔH) for forming lignin saturated solution could be calculated based on the eq. 3. The negative ΔH for all studied DESs systems (Table S6) indicated that the lignin dissolution in DESs was an enthalpy-driven process. Meanwhile, the Gibbs free energy change (ΔG) and the entropy change (ΔS) for forming lignin saturated solution could be calculated based on the lignin molar solubility (K) by the eqs. (4) and (5) [37]. The molar solubility could be estimated by assuming that the dissolution of lignin in DESs did not change the total volume of DESs. Thus, the molar solubility could be obtained by multiplying the DESs density by the molal solubility (Table S6).

$$\Delta H = H_{dis} \times m/n \tag{3}$$

$$\Delta G = -RTlnK \tag{4}$$

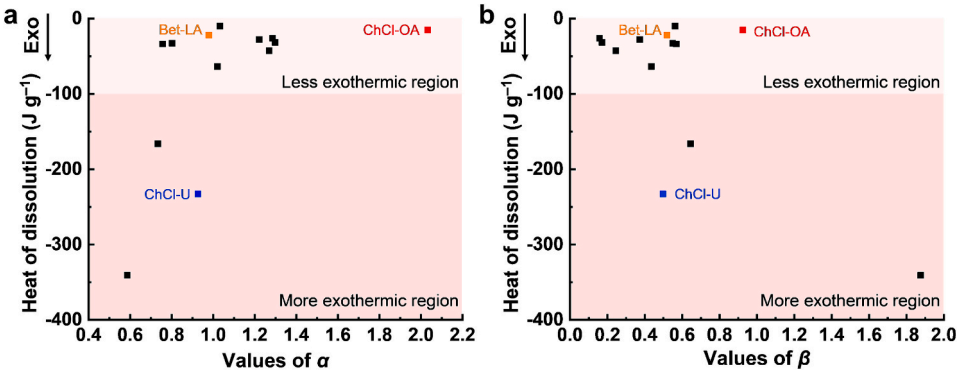


Fig. 5. Relationship between Kamlet-Taft parameters of DESs and dissolution heat of lignin.

$$T\Delta S = \Delta H - \Delta G \quad (5)$$

The n (moles of dissolved lignin) in eq. (3) was calculated based on the average molecular weight of the selected lignin (1759 g·mol^{-1}) (Fig. S5).

The values of ΔG were positive for all the present DES systems, which suggested that the interaction between the lignin and DESs might be weaker than that between HBAs and HBDs, and the process of lignin dissolution was non-spontaneous and unfavorable thermodynamically [38]. In addition, the negative ΔS and the relation $T\Delta S < \Delta H$ (eq. 5) were observed for lignin dissolution, suggesting that dissolution of lignin in DESs were enthalpically favorable and controlled by the entropy term. The negative ΔS could be generally affected by complex factors, *e.g.*, depolymerization, condensation, translation, rotation of solvated lignin macromolecular chains [30], and the interaction between lignin molecular chains and DESs molecules. As a refinement, ΔS could be divided into two components, *i.e.*, the mixing entropy change (ΔS_M) and the interaction entropy change (ΔS_I). The entropy changes before and after polymer dissolution calculated by the Flory-Huggins polymer theory only took into account the entropy change caused by the arrangement and movement of the molecular chains of lignin as they underwent depolymerization/condensation and translation/rotation [13,30], and did not take into account the entropy change caused by changes caused by the interaction between the chain segments and the solvents. Based on Flory-Huggins polymer theory equation, the ΔS_M could be calculated using the following equation: $\Delta S_M = -R[n_1 ln x_1 + n_2 ln(1 - x_1)] \div n_1$, where n_1, x_1, n_2, and R were the moles of dissolved lignin, the mole fraction of dissolved lignin, the moles of DESs, and the gas molar constant, respectively [39]. Because the mole fraction of dissolved lignin was between 0 and 1, $ln x_1$ and $ln(1 - x_1)$ were both negative and the ΔS_M was positive, favoring the mixing of lignin and DES. Therefore, the lignin dissolution in all the studied DESs were always associated with small entropy gained from the positive ΔS_M (Table S6), which was largely attributed to that the lignin macromolecular chains became more translatable and rotatable after being dissolved in DESs. The Flory-Huggins theory demonstrated that the driving force for dissolving lignin was the ΔS_M. Based on the discussion above, we hypothesized that lignin with an abundant aliphatic side chain structure and without a severely condensed aromatic ring backbone would favor the generation of positive ΔS_M, thus increasing the driving force for the dissolution of lignin in DESs. However, the ΔS was negative in all the studied DESs (Table S6), implying that the positive ΔS_M from depolymerization, condensation, translation and rotation of lignin macromolecular chains was counteracted to different extents by the negative ΔS_I, which strongly reduced the driving force of dissolving lignin in DESs. Moreover, the absolute values of ΔS_I were exponentially larger than the positive values of ΔS_M. Therefore, for the total entropy change of lignin dissolution in DESs, the ΔS_I played a predominant role, making the free energy change unfavorable for lignin dissolution. In comparison, the ΔS_M acted as a secondary role. The highly negative values of ΔS_I illustrated that the interaction between the dissolved lignin macromolecular chains and DESs molecules had lower entropy than the intramolecular freedom degrees of lignin macromolecular chains in the original amorphous state. During the dissolution of lignin in DESs, a huge amount of heat was released, which was beneficial for achieving spontaneous lignin dissolution. However, the robust interaction ability between DES molecules and lignin macromolecular chains lead to a significant reduction in interaction entropy and completely counteracting the increase in mixing entropy, which resulted in the $|T\Delta S|$ greater than $|\Delta H|$, and making the change in free energy unfavorable for lignin dissolution. Since a large number of DESs molecules strongly bounded to the polar and non-polar sites on the lignin chains, the freedom degrees of lignin macromolecular chains were more restricted, resulting in a stable and regular lignin-DESs solution [40]. These results were very similar with those for the dissolution of cellulose in ionic liquids [30]. Notably, the interaction entropy reduction for lignin dissolution in alkaline DESs showed much larger magnitudes than that for lignin dissolution in acidic and neutral DESs (Table S6), suggesting that the number of the alkaline DESs molecules bounded to the lignin macromolecular chains was much higher than that of acidic and neutral DESs molecules, and thus the lignin macromolecular chains were more firmly constricted in the alkaline DESs. As a result, the dissolution of lignin in DESs underwent from a confused amorphous state to the segregated chain disentanglement with extensive dynamic motion, followed by the formation of an ordered lignin solution with the lignin molecular chains firmly restricted by the DESs molecules, which could be explained in terms of ΔS_M and ΔS_I. Based on the above discussion, the compositions of DES played a key role in influencing the enthalpy changes and entropy changes of lignin solutions. Specifically, the HBA would assume a secondary role because of the large exothermic contribution of HBD to the system energetics. The dissolution of lignin by alkaline DES prepared from HBDs containing amino groups produced a large interaction entropy reduction and was violently exothermic, which was significantly larger than the enthalpy changes and entropy changes produced by acidic DES prepared from carboxylate group-containing HBDs.

Generally, it is necessary to controllably recover the dissolved lignin from the generated solution. Herein, all the dissolved lignin could be readily regenerated (Table S7). After correlating with the regenerated yield of lignin, it was observed that the lignin-DESs solution with larger ΔS_I values (< -800 J·mol^{-1}·K^{-1}) generally showed lower regenerated yield (Fig. 6 and Table S7). For instance, lignin dissolved in the alkaline ChCl-NMTU, ChCl-U, and ChCl-EA had much larger ΔS_I values (-896.14, -1279.14, and -1787.82 J·mol^{-1}·K^{-1}, respectively), and the lignin regeneration yields were all below 70 %. In comparison, lignin solutions (acidic and neutral DESs) with smaller ΔS_I values (> -400 J·mol^{-1}·K^{-1}) resulted in relatively higher lignin regenerated yield (> 85 %). The difference of regenerated yields implied that the stronger interaction between alkaline DESs molecules and the lignin molecular chains severely hindered the regeneration of lignin from the alkaline DESs solutions. From the interaction entropy of lignin solution, we could deduce that acidic and neutral DESs were the preferred solvents for the homogeneous modification of lignin to generate the target processed

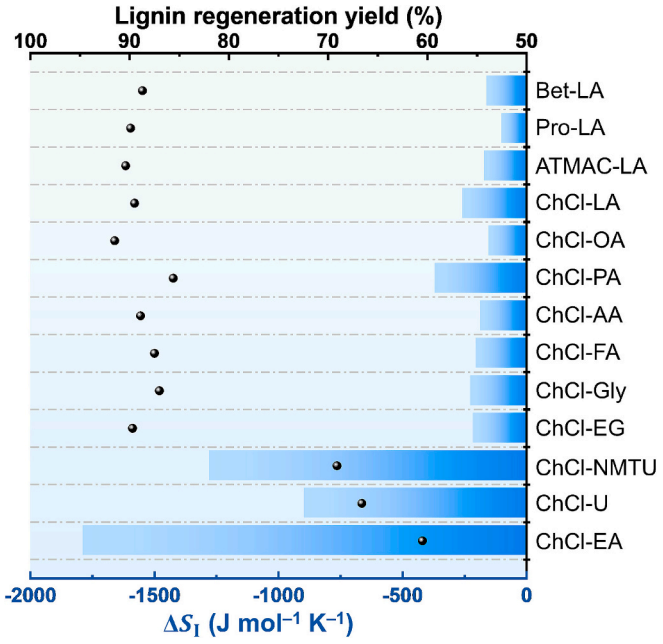


Fig. 6. Relationship between the ΔS_I values and the regenerated lignin yields. The blue bars and black dots represent the ΔS_I values and the regenerated lignin yields, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

lignin. Weaker interactions between the modified lignin macromolecular chains and the acidic, neutral DES molecules would be favorable for obtaining higher yields of the target lignin. In addition, acidic DES favored the protonation and dehydration of the hydroxyl group at α-carbon of lignin to generate reactive intermediates, which facilitated lignin functionalization through nucleophilic attack [41]. Meanwhile, lignin preserved a high content of aryl ether bonds and a large molecular weight after neutral DESs dissolution [2], the structure preservation promoted the preparation of naturally functionalized lignin.

4. Conclusions

In conclusion, the thermodynamic behaviors of lignin dissolution in alkaline, acidic, and neutral DESs were explored by determining lignin solubility and dissolution heat. It was discovered that dissolution of lignin in DESs was a violently exothermic process. Especially, the alkaline DESs showed a more vigorous exothermic process than acidic and neutral DESs during the lignin dissolution process. Moreover, the stronger interactions between the lignin macromolecular chains and DESs molecules at the dissolution state resulted in a highly negative entropy change, completely counteracting the positive entropy change from the net translation and rotation of the dissolved lignin chains. Therefore, lignin dissolution in DESs was enthalpically favorable and controlled by the entropy term, and the ΔS_I played the predominant role on making the free energy change unfavorable for lignin dissolution. Besides, the ΔS_I implied that lignin chains were more firmly constricted in the alkaline DESs than that in acidic and neutral DESs. These basic thermodynamic insights were highly helpful for designing innovative DESs to enhance the dissolution and even upgrading of lignin.

CRediT authorship contribution statement

Yang Wang: Writing – original draft, Methodology, Investigation, Conceptualization. **Huan Wang:** Investigation. **Chuanyu Yan:** Supervision, Methodology. **Weitao Wang:** Writing – review & editing, Supervision, Funding acquisition. **Bingzheng Jia:** Investigation. **Yiting Yao:** Investigation. **Lan Chen:** Investigation. **Qilu Hu:** Investigation. **Zhimin Xue:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

These authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this study.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijbiomac.2025.140224.

Data availability

Data will be made available on request.

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